

CT features of rounded atelectasis in chronic inflammatory pleural effusions in cats and dogs

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Abstract

Rounded atelectasis is well described in human medicine as focal lung deformation and collapse secondary to inflammatory pleural effusions and pleuritis. Specific CT features (round to ovoid soft tissue pulmonary attenuations, creation of an acute angle with the adjoining visceral pleura, and the presence of perinodular comet tail signs) support the diagnosis of rounded atelectasis in humans so that further diagnostic workup is not necessary in defining the nodules. In this retrospective case series, we described the CT characteristics of rounded atelectasis in eight cats and three dogs diagnosed with restrictive pleuritis secondary to either a chylothorax or pyothorax. Thirty-six soft tissue attenuating pulmonary nodular lesions were identified on CT. Comet tail signs, consisting of bundles of bronchi and vessels coalescing into the pulmonary nodules, were associated with 92% of the nodules (33/36), and 92% of the nodules abutted and created an acute angle with the pleura (33/36). Other prevalent features included location in gravity-dependent regions of the lung lobes (33/36, 92%), blurred hilar margins with sharper pleural margins of the nodules (33/36, 92%), presence of air bronchograms (30/36, 83%), homogeneous contrast-enhancement (23/36, 64%), and volume loss of the affected lung lobe (22/36, 61%). Pulmonary malignant neoplasms were not found cytologically (6/11 patients) or histologically (5/11 patients). To avoid a misdiagnosis of neoplasia, veterinary radiologists should be aware of the CT features of rounded atelectasis and consider it as a differential for pulmonary nodular lesions in patients with concurrent inflammatory pleural effusion and pleuritis.

KEYWORDS

atelectatic pseudotumor, Blesovsky syndrome, constrictive pleuritis, fibrosing pleuritis, folded lung

1 | INTRODUCTION

Rounded atelectasis (also called atelectatic pseudotumor, folded lung, or Blesovsky syndrome) is a condition extensively reported in human medicine, which describes focal, peripheral lung collapse caused by pleural adhesions and fibrosis secondary to chronic pleural effusions and pleuritis.^{1–15} This condition was first published in the 1960s by Blesovsky who reported that peripheral pulmonary nodules sampled during thoracotomies in humans frequently resembled benign folded

lung, which initiated research endeavors for defining the imaging characteristics, etiologies, and pathophysiology of this condition.^{1–3} Asbestos inhalation is the most documented factor associated with this condition in humans, although all types of pleural effusion can potentially induce it.^{1–15} Irritating pleural effusions (such as chylous or septic exudative effusions) have however a higher propensity of inducing restrictive pleuritis (also called fibrosing or constrictive pleuritis), by forming inelastic pleural fibrous tissue and adhesions between the parietal and visceral pleura, leading to decreased pulmonary

ventilatory capacity and ultimately the development of rounded atelectasis.^{16–23}

In human medicine, rounded atelectasis on CT has three main characteristics, with the lesions presenting as follows: (1) nodular pulmonary soft tissue attenuations, (2) abutting the visceral pleura and creating an acute angle with the pleural surface, and (3) having tortuous, compressed regional vessels and bronchi entering the nodule (called a comet tail sign).^{4–7,9–15} The pattern of contrast enhancement is documented to vary across cases.^{6,9–11,13,14} Multiple additional findings can further support the diagnosis, but may not always be present, and can include location in a gravity-dependent region, inclusion of air bronchograms, volume loss of the affected lung lobe, and thickening of the adjacent pleura, among other signs.^{4–7,9–13,15}

Ante-mortem identification of such rounded pulmonary lesions in cats and dogs diagnosed with a chronic irritative pleural effusion and pleuritis could lead to an incorrect diagnosis of pulmonary neoplasia. Our aim is therefore to describe and characterize the CT features of rounded atelectasis in cats and dogs diagnosed with restrictive pleuritis and chylothorax or pyothorax.

2 | METHODS

2.1 | Selection and description of subjects

The study was a retrospective, descriptive, single-center, case series design. As this was a retrospective study and the images were acquired for clinical purposes, an animal care and use protocol was not required. Owner consent was obtained for all diagnostics, treatments, and use of acquired data at the time of patient presentation. The use of the data included in the study was approved by the hospital administration.

Medical records and imaging studies of cats and dogs presented at our institution between August 2012 and February 2022 were searched to identify patients with the following criteria: (1) confirmation of a chylothorax or pyothorax diagnosis via pleural fluid analysis by an American College of Veterinary Pathology (ACVP)-certified veterinary clinical pathologist and (2) a thoracic CT examination (pre- and post-intravenous contrast medium administration) performed following the chylothorax or pyothorax diagnosis. A total of 75 chylothorax cases and 43 pyothorax cases were identified based on these criteria. Patients were included in the study if: (a) they had CT findings consistent with restrictive pleuritis, (b) pulmonary nodular lesions were identified on CT, and (c) fine needle aspiration, biopsy, or autopsy with cytological or histological analysis of the pulmonary nodular lesions by an ACVP-certified veterinary pathologist (clinical pathology and/or anatomic pathology (T.A.D., H.D.) were performed within 2 weeks of the CT examination and had no evidence of malignancy. All fluid analysis, cytology, histopathology, and gross autopsy evaluations were available at the time the CT images were reviewed for this study. The determination of restrictive pleuritis used in this study was based on the documented criteria from Suess et al.,¹⁸ in that inclusion of 2 out

of the 3 following imaging features were consistent with restrictive pleuritis: rounding of the lung lobes, persistent retraction of the lung lobes, and/or thickening of the pleura.

Final decisions for subject inclusion were made by a rotating veterinary surgery and medicine intern (D.T.H.T., currently first-year radiology resident) and a board-certified veterinary radiologist (A.L.R., European College of Veterinary Diagnostic Imaging [ECVDI] and American College of Veterinary Radiology [ACVR]).

2.2 | Data recording and analysis

Subject signalment, diagnostic data, and long-term outcome information were retrieved from medical records by the rotating veterinary surgery and medicine intern (D.T.H.T.). The CT examinations were evaluated by both the intern (D.T.H.T.) and the ECVDI/ACVR-certified veterinary radiologist (A.L.R.), who reached a consensus on each lesion, after they reviewed the human literature and familiarized themselves with the disease process and its CT imaging features. Observers were aware of all medical record information at the time of image interpretation. Each pulmonary nodular lesion was evaluated for shape (round, ovoid, or other), attenuation (soft tissue or other), margination (well-defined or ill-defined), consistency (homogeneity or heterogeneity before and after intravenous contrast administration), presence of air bronchograms, lung lobe involvement, and location within the parenchyma (including if the nodule contacted the visceral pleura in any plane).

The angle between a pulmonary nodular lesion and the contacting visceral pleura was characterized as acute or obtuse, based on its assessment in multiple plans using tangential rays created from points immediately adjacent to the point of contact on the associated nodular and pleural surfaces (Figure 1A). The presence of comet tail signs was determined by comparison to CT imaging examples from human medical literature^{4–6,10–11,13,15} and by use of the common definition reported across human medical literature: curving and compressed vessels and bronchi converging towards the nodule.^{5–6,9–12,15}

3 | RESULTS

Eleven patients, consisting of eight cats and three dogs, met the inclusion criteria and were included in this study. Nine patients had a chylothorax and two patients had a pyothorax. Of the 107 patients excluded, 42 did not have any evidence of restrictive pleuritis or pulmonary nodules on CT examination. Seventeen patients had evidence of restrictive pleuritis but no evidence of pulmonary nodules on CT examination. Of the remaining patients with pulmonary nodules, eight had no evidence of restrictive pleuritis while 32 patients had no cytology or histology performed. Of the remaining patients with the nodules sampled, six were determined to be carcinoma, one was consistent with an abscess, and one was consistent with lung lobe torsion.

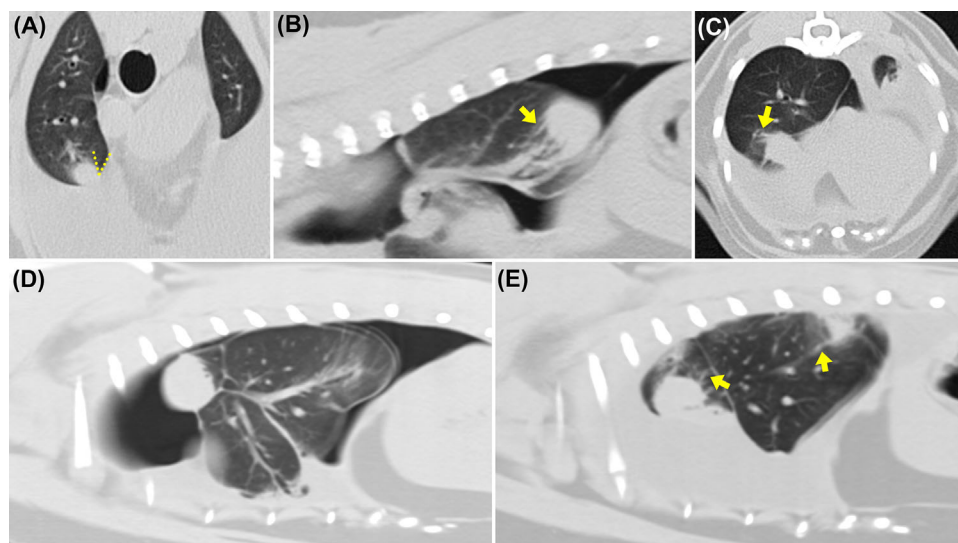


FIGURE 1 Pre-contrast thoracic CT transverse (A, C) and multiplanar reconstructions (MPR) parasagittal (B, D, E) images in a lung algorithm in dogs (A, C) and cats (B, D, E) diagnosed with chylothorax and restrictive pleuritis. Perilesional tortuous bronchi and vessels coalescing to the pulmonary nodular lesions (comet tail signs) are present (yellow arrows on images B, C, E). The pulmonary nodular lesions additionally create an acute angle with the abutting visceral pleura, depicted by the discontinuous yellow lines on image A. Rounding of the lung margins and a moderate pneumothorax (B, D) can also be identified [Color figure can be viewed at wileyonlinelibrary.com]

3.1 | Clinical presentation

The age of the eight included cats ranged from 2 to 11 years, with a mean age of 4.8 years. There were two spayed female and 6 castrated male cats. Six were Domestic Shorthairs, one was an American Shorthair, and one was an Abyssinian. The age of the three included dogs ranged from 3 to 10 years, with a mean age of 5.6 years. All dogs were male, and two were castrated while one was intact. Two were Labrador Retrievers and one was a Shiba Inu. All patients presented with dyspnea ($n = 11$), with or without tachypnea ($n = 9$). Other clinical signs included decreased lung sounds often localized to the ventral lung fields ($n = 8$), increased or harsh lung sounds often localized to the dorsal lung fields ($n = 4$), dehydration ($n = 3$), muffled heart sounds ($n = 3$), and muscle wasting or weight loss ($n = 3$).

3.2 | Computed tomography

Thoracic computed tomography was performed on all 11 patients under general anesthesia with a third-generation CT Toshiba Aquilion 64 unit (Canon Medical Systems, Tustin, CA, USA). Patients were placed in sternal recumbency and nonionic iodinated contrast (2.2 mL/kg; Omnipaque 300, GE Healthcare, Princeton, NJ, USA) was administered intravenously after the acquisition of the initial precontrast images. An arterial phase of the thorax was acquired in three patients, in addition to a standard venous phase. Images were acquired at 120 kVp and 100 mAs, using a 512×512 matrix, with a 3 mm slice thickness, and a beam pitch of 1, using soft-tissue, lung, and bone algorithms.

The CT findings are summarized in Table 1. The examinations of all 11 patients were consistent with restrictive pleuritis, with

contrast-enhancing pleural thickening and retraction and rounding/scalloping of the lung lobe margins (Figures 1–5). All 11 patients had evidence of pleural effusion in varying amounts characterized by low Hounsfield units (HU) ranging approximately from 0 to 20 HU for the patients with chylothorax (Figures 1–5) and 0 to 50 HU for the patients with pyothorax. Pneumothorax was seen in 10 of 11 patients, all post-thoracocentesis or thoracostomy tube placement, and therefore likely iatrogenic (Figures 1–4). Sternal lymphadenopathy was present in all patients.

All 11 patients had multiple, round to ovoid soft tissue attenuating nodules distributed throughout multiple lung lobes (Figures 1–5). The number of nodules varied from two to five in any one patient, totaling 36 nodules among the 11 included patients. There was no specific lung lobe predilection. Most of the nodules (33/36, 92%) had at least one margin abutting and creating an acute angle with the visceral pleura, and most nodules (33/36, 92%) also had evidence of curvilinear vessels and bronchi tapering to the nodule (comet tail sign) (Figures 1–3 and 5). The 3 nodules lacking these latter two CT features occupied the entire lung lobe, precluding the visualization of perilesional comet tails or the formation of an acute angle with the visceral pleura (Figure 4).

Contrast enhancement was present in all nodules (36/36, 100%) but the pattern of enhancement varied, with most (23/36, 64%) homogeneously contrast-enhancing (Figures 3C, 4C, and 5B). The remaining nodules (13/36, 36%) were heterogeneously contrast-enhancing, specifically with a hyperattenuating outer margin and hypoattenuating center (Figure 3F). Additional findings included location in gravity-dependent regions of the lung lobes (33/36, 92%), blurred hilar margins with sharper pleural margins (33/36, 92%, Figure 2), incorporation of at least one air bronchogram in the central or peripheral regions of

TABLE 1 CT features of rounded atelectasis in 8 cats and 3 dogs

Primary features	Numbers (%)	Additional features	Numbers (%)
Pulmonary nodular soft tissue attenuation	36/36 (100%)	Located in gravity-dependent regions of the lung lobes	33/36 (92%)
Abutting and creating an acute angle with the visceral pleura	33/36 (92%)	Blurred margins in the hilar aspect and more defined margins in the pleural aspect	33/36 (92%)
Comet tail signs	33/36 (92%)	Air bronchogram in the center or peripheral region	30/36 (83%)
		Volume loss of the affected lung lobe	22/36 (61%)
		Hypoattenuation of the adjacent parenchyma in the affected lobe	14/36 (39%)
		Post-contrast higher attenuation at the periphery with a more hypoattenuating center	13/36 (36%)

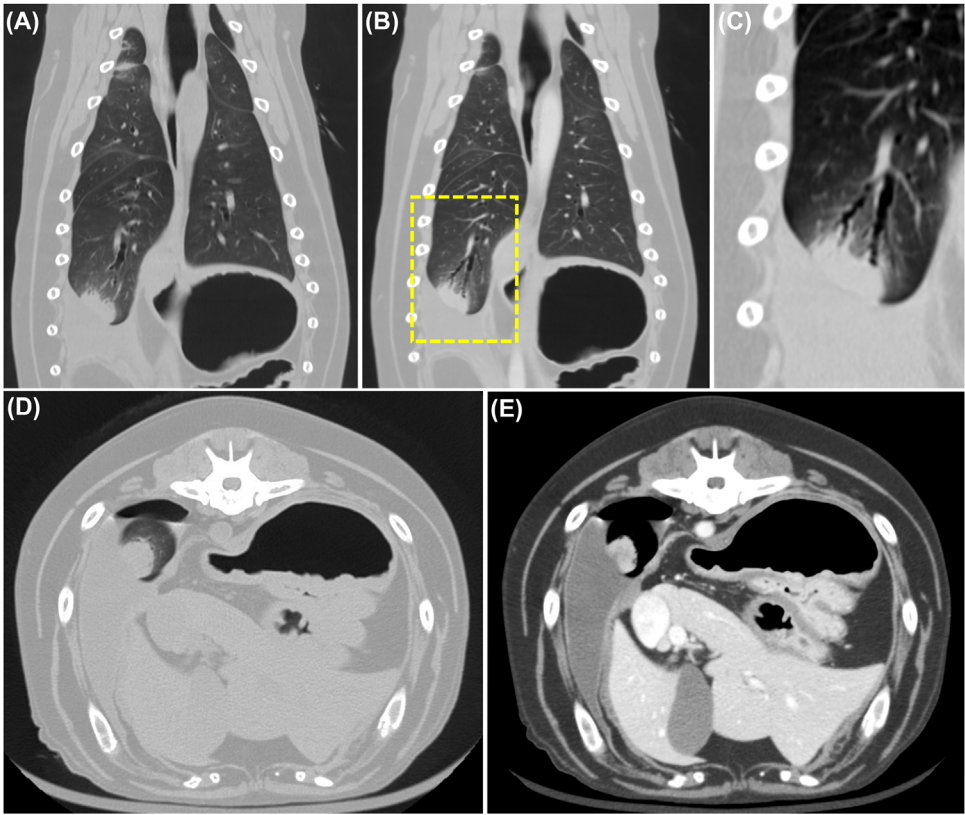


FIGURE 2 Pre-contrast dorsal MPR (A, B, C) and transverse (D) images in a lung algorithm, and post-contrast (E) transverse image in a soft tissue algorithm in a dog diagnosed with a chylothorax and restrictive pleuritis. Image C is a cropped image of the nodular lesion in image B. A soft tissue pulmonary nodular lesion is present in the right caudal lung lobe, abutting the visceral pleura with an acute angle. This nodule has blurred hilar margins compared to its pleural margins, with multiple small air bronchograms. A comet tail sign can be noted on the dorsal images, characterized by tortuous perilesional bronchi and small vessels. A small-volume, right-sided pneumothorax is observed on image D and right-sided pleural effusion can be identified on image E with mild associated pleuritis [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/terms-and-conditions)]

the nodule (30/36, 83%, Figures 2, 3), associated volume loss of the affected lung lobe (22/36, 61%), and hypoattenuation of the surrounding lobar pulmonary parenchyma (4/36, 39%). Contrary to what has been reported in humans, focal, perinodular visceral pleural thickening was not identified in our patients.

3.3 | Cytology, gross autopsy, and histopathology findings

Ultrasound-guided fine needle aspirations for cytology of the pulmonary nodular lesions were performed in six patients (1 dog, 5 cats),

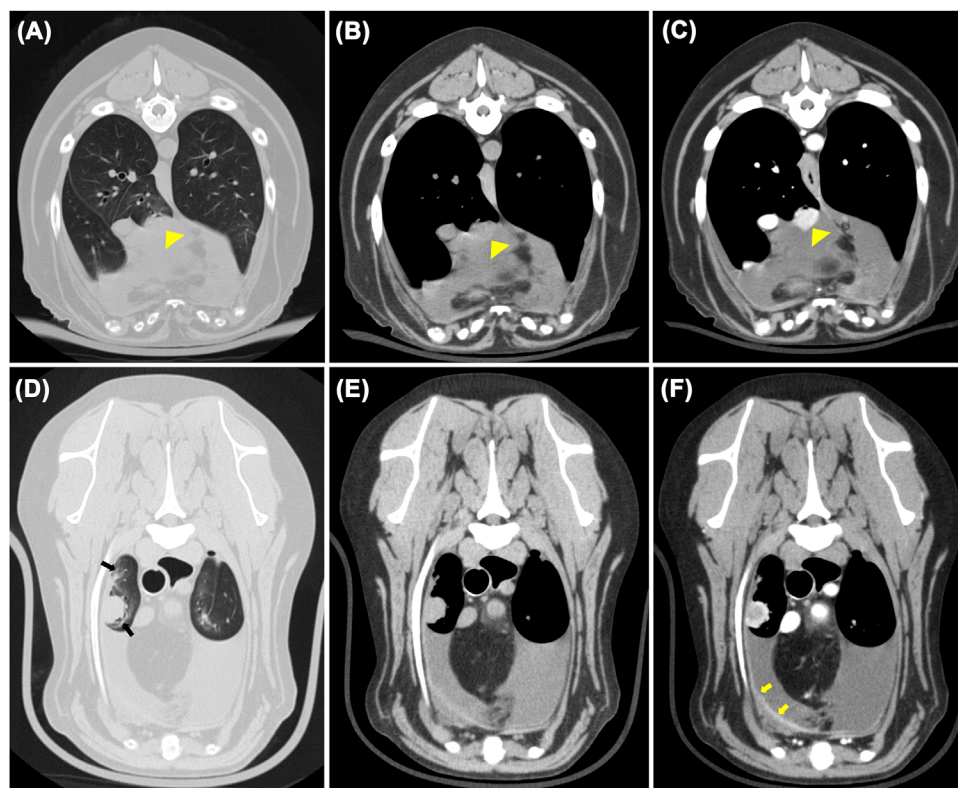


FIGURE 3 Pre-contrast transverse thoracic CT images in a lung algorithm (A, D), and pre- (B, E) and post-contrast images (C, F) in a soft-tissue algorithm of a dog diagnosed with chylothorax and restrictive pleuritis. Two soft tissue pulmonary nodular lesions can be identified, one in the ventral aspect of the accessory lung lobe (yellow arrowheads, images A, B, C), and one in the right cranial lung lobe (D, E, F). Comet tail signs with coalescing vasculature (black arrows, image D) are present around the right cranial lung lobe nodular lesion. The nodule in the accessory lung lobe homogeneously enhances and has a minuscule air bronchogram at its dorsal aspect. The nodule in the right cranial lung lobe heterogeneously enhances in its periphery, while its center remains hypodense. Parietal pleural thickening (small yellow arrows, image F), with bilateral pleural effusion, and a scant volume of pneumothorax (image D, left-sided) can also be noted [Color figure can be viewed at wileyonlinelibrary.com]

all with chylothorax. Surgical biopsies were performed in one cat that was diagnosed with pyothorax. Autopsies were performed in four patients (2 dogs, 2 cats).

The cytology results in three of six patients that underwent aspiration of the pulmonary nodular lesions were consistent with chylous fluid. In the remaining three patients, mixed inflammation, either predominantly suppurative ($n = 2$) or histiocytic ($n = 1$), was found. None of the cytological results revealed infectious agents or neoplastic cells.

The biopsy results in the one patient (1 cat) that underwent surgical biopsies had evidence of chronic, proliferative, and pyogranulomatous pleuritis consistent with pyothorax and simultaneous marked atelectasis. The pleura was diffusely and severely expanded by mixed inflammation (neutrophils, epithelioid macrophages, lymphocytes, plasma cells, and eosinophils), vascularized fibrous (granulation) tissue, hemorrhage, edema, and necrotic debris. Pleural mesothelial cells were hypertrophied and hyperplastic. Within the pulmonary parenchyma, eosinophilic alveolitis, bronchitis with bronchiectasis, smooth muscle hypertrophy, and glandular hyperplasia were reported (interpreted to represent features of feline asthma).

In all four patients (2 cats and 2 dogs) with autopsies, chronic pleuritis with fibrosis (restrictive pleuritis) corresponded with grossly identified shrunken, rounded lung lobes (Figure 4E). In one of the dogs,

dissection of the nodular pulmonary lesions revealed contracted and folded pulmonary parenchyma with scant intralesional chylous effusion. A subgross view of the lung lobe from the other dog showed compressed, converging bronchioles and arteries at the border of the atelectatic and normal parenchyma, likely corresponding with the comet tail sign in the CT images (Figure 6). Similarly in one cat that underwent autopsy, the apical aspect of the left caudal lung lobe revealed a focally extensive collapsed portion of pulmonary parenchyma where the blood vessels were brought closer in apposition to each other, interpreted to correspond with the comet tail sign identified on CT. There was no evidence of pulmonary neoplasia or infectious agents. Additional thoracic lesions included pulmonary microthrombosis, subpleural pneumonia, pericarditis with pericardial atypical mesothelial proliferation, bronchiolitis and pulmonary interstitial fibrosis, acute apical cardiac herniation through the pericardial sac, with acute fibrinohemorrhagic epicarditis and cardiomyocyte necrosis, free-floating intrathoracic necrotic material with peripheral fibrosis, diaphragmatic myositis and fibrinoid vascular necrosis, chronic epicarditis, pleural lymphoid follicle formation with accumulation of hemosiderin and lipofuscin pigments, mediastinitis, and mediastinal lymph node hyperplasia. Mesothelial hypertrophy and hyperplasia of the mediastinal structures were identified in all cases.

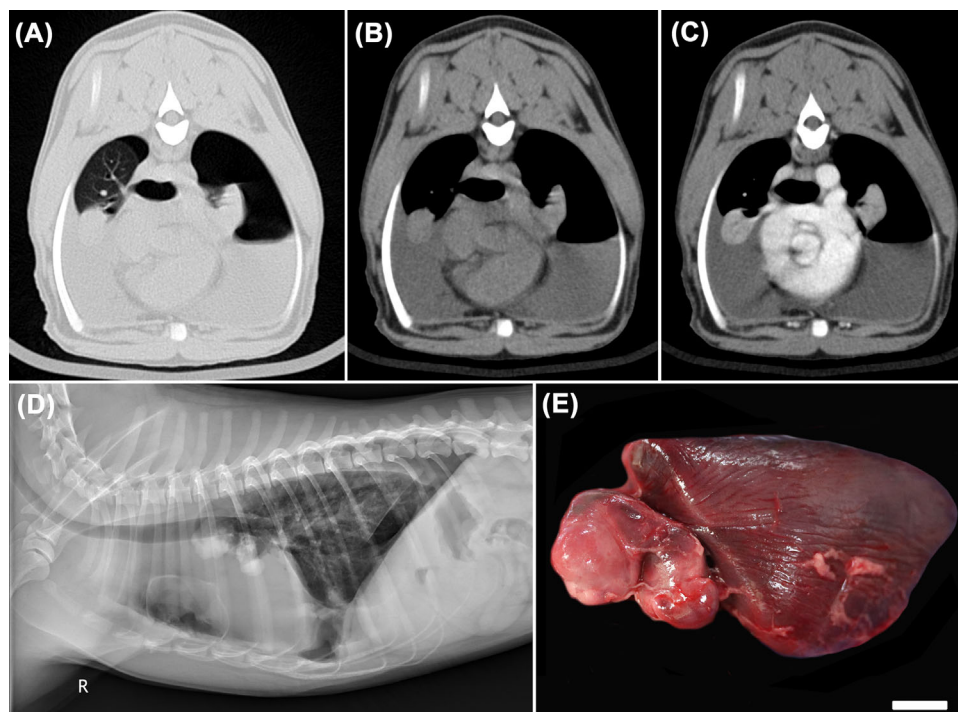


FIGURE 4 Transverse pre-contrast thoracic CT images in (A) lung algorithm and (B) soft tissue algorithm, and post-contrast thoracic CT image (C) in a soft tissue algorithm in a dog diagnosed with chylothorax and restrictive pleuritis. There are bilateral nodular lesions noted in the ventral aspect of the caudal lung lobes with decreased lung volume (especially on the left) and rounding of the pulmonary margins. A bilateral pneumothorax (A) and pleural effusion (B, C) are also present. Image D is a corresponding right lateral thoracic radiograph of the same dog showing severe nodular atelectasis of the cranial and caudal subsegments of the left cranial lung lobe, along with a mild pneumothorax and minimal pleural effusion. Image E is a macroscopic picture of the left lung lobes, depicting the severe nodular atelectasis of the left cranial lung lobe with multiple fibrous adhesions seen between all lung lobes. Bar is 1.0 cm

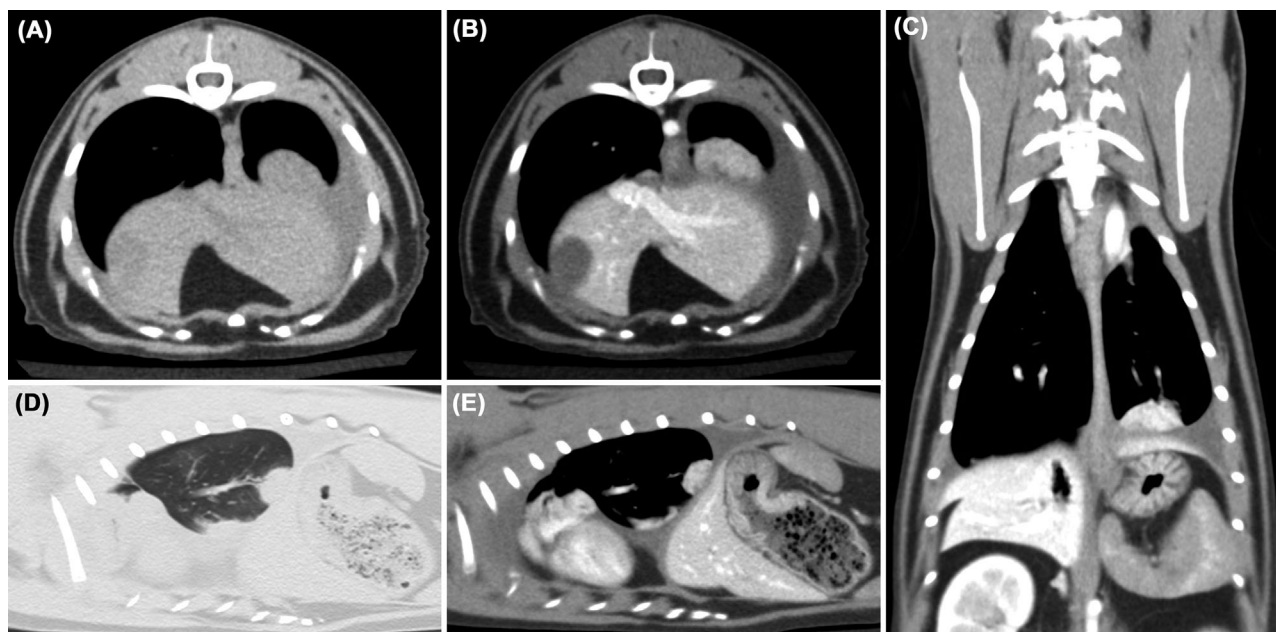


FIGURE 5 Transverse pre- (A) and post-contrast (B) and MPR post-contrast (C, E) thoracic CT images in a soft tissue algorithm of a cat diagnosed with a chylothorax and restrictive pleuritis. Image D is a pre-contrast MPR parasagittal image of the same cat in a lung algorithm. A left caudal lung lobe nodular lesion is present, abutting the visceral pleura with an acute angle (A, D), with more irregular cranial margins, and thin tortuous bronchi and vessels identified in the periphery of this lesion (D). Mild bilateral pleural effusion and rounding of the pulmonary margins are also present on all images

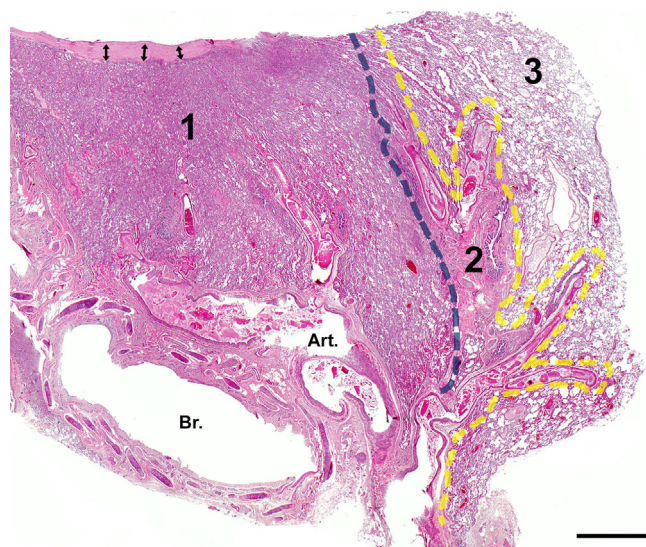


FIGURE 6 Subgross image of lung from a dog diagnosed with chylothorax and restrictive pleuritis. Compressed, atelectatic pulmonary parenchyma is represented by 1. Thickened visceral pleura expanded by eosinophilic connective tissue (fibrosis) is denoted by the arrows. A large bronchus (Br.) and Artery (Art.) are found at the bottom of the image. At the junction between the atelectatic (1) and normal (3) pulmonary parenchyma (2, dotted grey line), compressed, converging arteries and bronchioles extend into the regional parenchyma (dotted yellow lines), corresponding with the comet tail sign on the CT imaging. Bar is 1.0 mm. Hematoxylin and Eosin

3.4 | Clinical outcome

Treatment protocols among the patients varied but included supportive care in four of nine chylothorax patients (2 dogs and 2 cats), surgical management in two of nine chylothorax (1 dog and 1 cat) and both pyothorax patients (2 cats), and no treatments initiated in three of nine chylothorax patients (3 cats). Supportive care in these cases included initiating various combinations of rutin, low-fat diets, corticosteroids, and antimicrobials. Of the four medically managed patients, one dog was euthanized 9 days after presentation, one cat survived 4 years after the presentation and is still alive, and the remaining two patients were lost in follow-up immediately after discharge. Surgical intervention of the two chylothorax patients (1 cat and 1 dog) included thoracoscopic thoracic duct ligation, subtotal pericardiectomy, Pleural-Port (Norfolk Vet Products, Skokie, IL) placement, and intraoperative pericardial and pleural biopsies; the dog additionally underwent cisterna chyli ablation. Of these two patients, the dog was euthanized the same day as surgery for post-operative decompensation, and the cat arrested and died in the hospital 3 days post-operatively and 22 days after initial presentation. The two pyothorax patients (2 cats) underwent surgical intervention, which included median sternotomy with surgical exploration with additional lobar lung lobectomy for one of the patients. Of these two patients, one was euthanized 4 days post-operatively due to progressive severe restrictive pleuritis and the other was reported to be doing well 2 weeks post-operatively. Three chylothorax patients (3 cats) had no treatments initiated; one was

lost in follow-up immediately after discharge, one was reportedly doing well 2 months after discharge but has since been lost in follow-up, and the last patient was euthanized without initiation of any treatments.

4 | DISCUSSION

Based on our results, rounded atelectasis should be included as a differential diagnosis for pulmonary nodular lesions identified on CT examinations of cats and dogs diagnosed with restrictive pleuritis and either chylothorax or pyothorax. In our study, 33 of 36 nodules (92%) met the CT criteria described in human medicine for rounded atelectasis: (1) round or ovoid soft tissue attenuations within the pulmonary parenchyma, (2) creation of an acute angle with the visceral pleura, and (3) the presence of perinodular comet tail signs. Three nodules occupied the entirety of a lung lobe, leading to a lack of visualization of the comet tail sign and an acute angle between these nodular lesions and the pleura.

Two main theories currently exist to explain the formation of rounded atelectasis in human medicine. The “folding (pleural effusion)” theory suggests that pleural effusion compresses the adjacent lung so that it becomes atelectatic and concentrically folds in on itself with pleural effusion entrapped in the invaginated portion of the pulmonary parenchyma.^{6,8,11–15} The “fibrosing” theory suggests that pleural injury, mostly due to asbestosis in humans, leads to pleural fibrosis and later contraction of the affected pleura and adjacent pulmonary parenchyma.^{6,8,11–15} In both theories, fibrous adhesions ultimately bridge the overlying pleural surfaces, leading to formation of nodule-like lesions of the atelectatic portion of the lung.^{6,8,10–15} In this study population, dissection of one of the pulmonary nodules in one patient revealed intralesional pleural effusion within the contracted and folded pulmonary parenchyma, which may support the folding (pleural effusion) theory. The amount of intralesional effusion in this patient was however scant and could have represented post-mortem translocation of pleural fluid. Similarly, the cytological results revealed chylous fluid aspiration in three of six patients that underwent fine needle aspirations of their pulmonary nodular lesions, also possibly supporting the folding (pleural effusion) theory, with trapped intralesional chylous fluid. However, sampling of the pleural effusion while reaching for the pulmonary nodules cannot be completely excluded in these cases.

It is of tremendous prognostic importance to be able to differentiate rounded atelectasis from malignant neoplasia on CT images. In human medicine, the most sensitive and specific CT signs of rounded atelectasis are the presence of a round or ovoid soft tissue attenuation within the pulmonary parenchyma, peripheral location with creation of an acute angle with the visceral pleura, and presence of perinodular comet tail signs.⁷ In veterinary medicine, pulmonary malignant neoplasms tend to occur at the periphery of the lung lobes and because of their later stage diagnoses and the relatively smaller patient size in comparison to humans, these neoplasms usually have a contacting margin with the adjacent visceral pleura.^{24,25} The type of pleural contact may

however differ between rounded atelectasis and neoplasms in veterinary medicine. A sharp, acute angle with the pleura is present with rounded atelectasis, whereas pleural contact in neoplasia can have a more variable appearance, with the occasional presence of pleural tags (also called pleural tail signs), which is not present in rounded atelectasis in humans. Pleural tags are described as linear soft tissue attenuations bridging the tumor to the adjacent pleura and caused by septal thickening secondary to lymphatic obstruction, edema, or malignant spread.^{7,24–26} Pulmonary neoplasms may also contain foci of intralesional gas or mineralization with local invasion into the bronchi, blood vessels, or adjacent lung lobes, which were all features not identified in our study with rounded atelectasis.^{13,24,25} In human medical literature, the pattern of contrast-enhancement described with rounded atelectasis also varies,^{6,7,9–11,13,14} although in our study, the majority of the nodules were homogeneously contrast-enhancing. This pattern of enhancement is however nonspecific and would not on its own allow to differentiate between rounded atelectasis and neoplastic nodules.

Several limitations were present in this study, including an inherent sampling bias due to the retrospective selection of cases, and the small sample size preventing statistical comparison. Another limitation was the low number of histological evaluations performed (5/11 patients). While neoplasia was not found cytologically after fine needle aspirations for the remaining six of 11 patients, a poorly exfoliating pulmonary neoplasm could have confounded the most accurate diagnosis in these cases. Additionally, due to the inherent risks associated with lung sampling in patients diagnosed with restrictive pleuritis, all the pulmonary nodular lesions detected on CT examination could not be assessed cytologically. The decision of which nodules to aspirate under ultrasonographic guidance was made based on their CT appearance and representability in comparison to the unreachable pulmonary lesions, as well as feasibility and safety of the procedure. Therefore, even though each nodule was not assessed cytologically, each of these six of 11 patients had at least one representative pulmonary nodular lesion cytologically sampled, which shared similar CT features with the other, more difficult to reach, pulmonary nodules, as well as histologically confirmed rounded atelectasis nodules and widely described rounded atelectatic nodules in humans.^{4–7,9–15} Finally, only two of six of the patients that underwent sampling of their pulmonary lesions and were not euthanized or died in the hospital had adequate follow-up information after discharge. One of these patients has been doing well without evidence of progressive disease 4 years after the initial CT examination and sampling of the pulmonary nodular lesions, which further supports a benign process, while the follow-up information available for the other case encompasses a short time frame (2 weeks post-operatively) since retrospectively searching for cases for this study. Of the remaining patients, four patients were lost in follow-up soon after the CT scan and sampling were performed, while the last five patients were euthanized or died in the hospital shortly after surgery or discharge from the hospital.

In conclusion, the presence of nodular lesions in the periphery of the lungs in patients diagnosed with chylothorax or pyothorax and

with CT findings suggestive of restrictive pleuritis should prompt clinicians to consider rounded atelectasis as a differential diagnosis. A large percentage of the cases in this series (92%) exhibited characteristic features of rounded atelectasis described in humans, with the presence of an acute angle with the adjoining visceral pleura and comet tail signs. The lack of these features in a small percentage of the cases was attributed to the distribution of some of the lesions and occupation of an entire shrunken lung lobe, therefore precluding their visualization. It should be noted that this study was limited to patients with restrictive pleuritis secondary to chylothorax and pyothorax, but rounded atelectasis may also be present with any condition leading to pleural effusion, inflammation, and fibrosis. Future studies should aim to confirm the presence of rounded atelectasis in different patient populations and comparing these CT characteristics with pulmonary neoplasms.

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LIST OF AUTHOR CONTRIBUTIONS

Category 1

- a. Conception and Design: Tran, Le Roux
- b. Acquisition of Data: Tran, Donovan, Daverio, Le Roux
- c. Analysis and Interpretation of Data: Tran, Donovan, Daverio, Le Roux

Category 2

- a. Drafting the Article: Tran
- b. Revising Article for Intellectual Content: Tran, Donovan, Daverio, Le Roux

Category 3

- a. Final Approval of the Completed Article: Tran, Donovan, Daverio, Le Roux

Category 4

- a. Agreement to be accountable for all aspects of the work in ensuring that questions related to the accuracy or integrity of any part of the work are appropriately investigated and resolved: Tran, Donovan, Daverio, Le Roux

CONFLICT OF INTEREST STATEMENT

The authors declare no conflicts of interest or relationship, financial or otherwise, that might be perceived as influencing their objectivity in the reporting of this study.

PREVIOUS PRESENTATION OR PUBLICATION DISCLOSURE

The results of this study were presented as an oral presentation at the ACVR annual meeting in Reno, NV on October 20, 2022.

REPORTING CHECKLIST DISCLOSURE

STROBE-VET Reporting Guidelines were followed in this study.

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